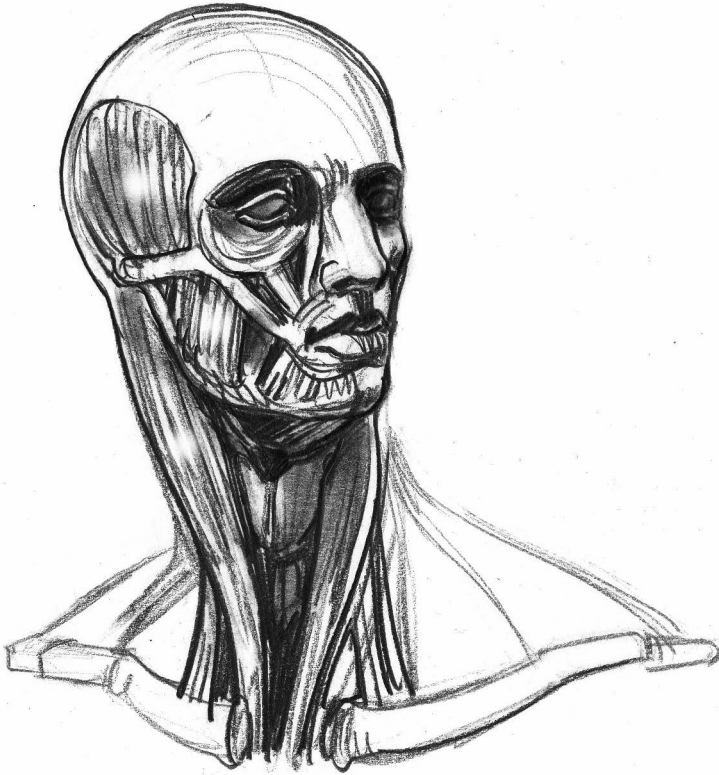


An introduction to

morpho

anatomy drawing basics

Michel **Lauricella**

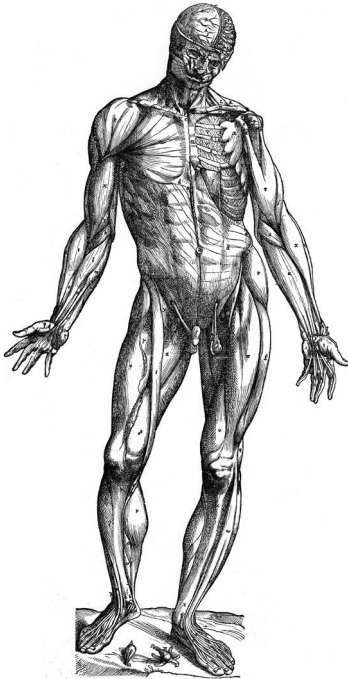


rocky**nook**

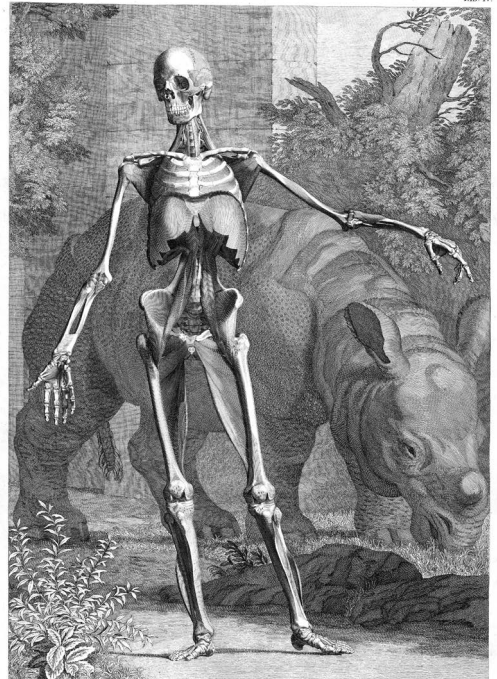
These skeletons or écorchés stupefy because they act like living creatures.

—Roger Caillois, *Au coeur du fantastique* (At the Heart of the Fantastic), Gallimard, Paris, 1965.

The material in this ebook is excerpted from *Morpho: Anatomy for Artists* (978-1-68198-374-5), by Michel Lauricella. © 2014 Groupe Eyrolles, Paris, France. Translation © 2018 Rocky Nook, Inc.



André Vésale (1514–1564) and Jan Steven Van Calcar (1499–1546), *L'Epitome* (*The Epitome*), 1543.



Bernhard Siegfried Albinus (1697–1770) and Jan Wandelaar (1690–1759), *Tabulae Sceleti et Musculoru Corporis Humani* (1747).

The Écorché: a Genre

During the Renaissance, artists began producing works depicting the anatomy of the human body. These works were intended for art lovers, as well as for use by physicians. Given that Leonardo da Vinci (1452–1519) never finished his treatise, the 16th century seven-volume set of illustrated books, *La Fabrica*, by André Vésale is considered as marking the beginning of a long tradition of anatomical drawing that endures to this day.

Throughout history, the care that has been given to the representation of écorché figures, originally intended as simple anatomical studies, has made them into a subject in themselves—a genre just like the nude or the landscape. This genre also has its own history, codes, and conventions, and can be used as a means of personal expression. These écorché characters captivate us, with their bodies



Jacques Fabien Gautier d'Agoty (1716–1785), *Complete Myology, in Natural Color and Size* (1746) (renamed *Anatomical Angel* by the surrealists).

stripped and laid bare, improbably suspended between life and death. The surrealists later appreciated their strongly fantastical aspect.

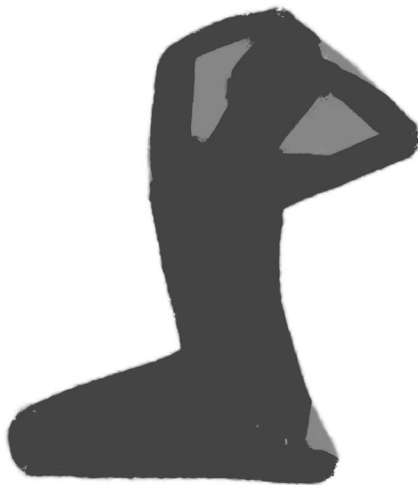
Morphology

As early as 1890, Paul Richer preferred using the term “morphology” to “anatomy,” as he believed the term “morphology” was more about aesthetics and the form as a whole rather than about medicine and its individual details. We shorten it “morpho.”

This morpho approach involves retaining only those elements of anatomy that determine shape (we simplify and merge certain groups of muscles as necessary), making

whichever anatomical element is most prominent under the skin coincide with the outlines of the drawing. In other words, the thickness of the skin is no longer taken into account and, depending on the regions of the body and the morphological characteristics of your model, you might choose to use an element of bone or muscle or fat as the underlying element that gives structure to the form.

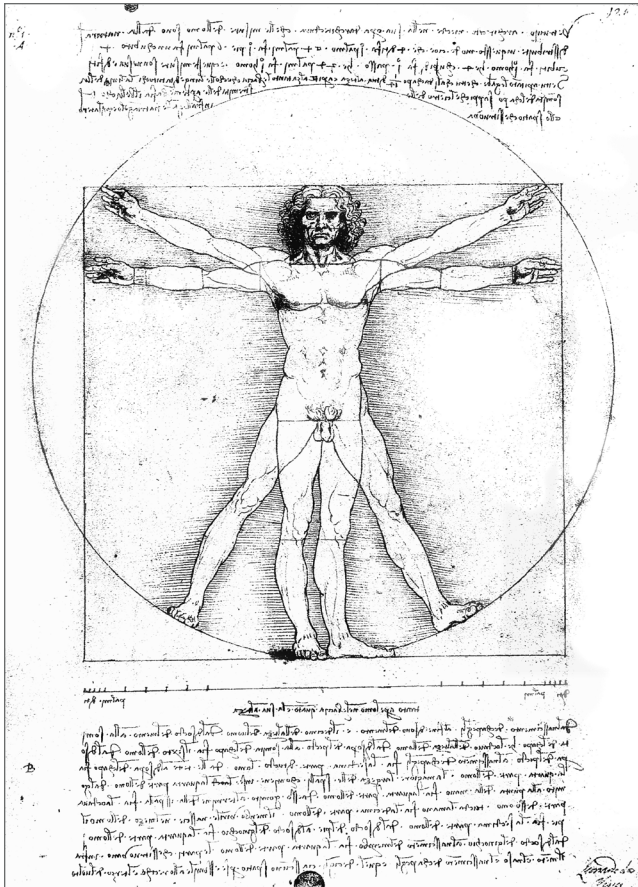
We will, in fact, give equal importance to fat, though we will define its shape in a somewhat arbitrary manner, because unlike bones and muscles, fat develops underneath the skin and has no clear boundaries. I have included some sketches to demonstrate the drawing of fat.



Drawing the Écorché

An écorché drawing can be produced in several stages. First, I suggest you draw the basic composition of your figure using simple geometric shapes to create the overall silhouette. Check the proportions by measuring the various body parts and compare them to each other. Make sure your vertical lines are true by comparing the silhouette of the model with the vertical lines of the architectural space (assuming you have no actual plumb line) and the boundaries of your frame.

Now it is time to produce the écorché. It is helpful to note on your drawing all of the bone reference points, to graphically distinguish the hard and soft areas. Then connect these reference points beginning with the largest elements, such as the rib cage (with a simple shape like that of an egg), the pelvis (like a large box of matches), and the skull. The orientation of these first elements is essential to express the dynamics of a pose.



Leonardo da Vinci,
Vitruvian Man, c. 1490

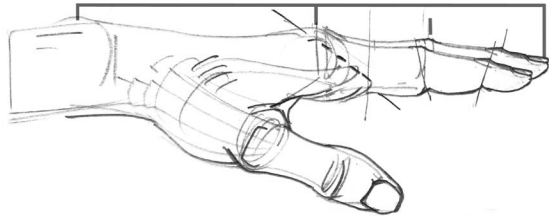
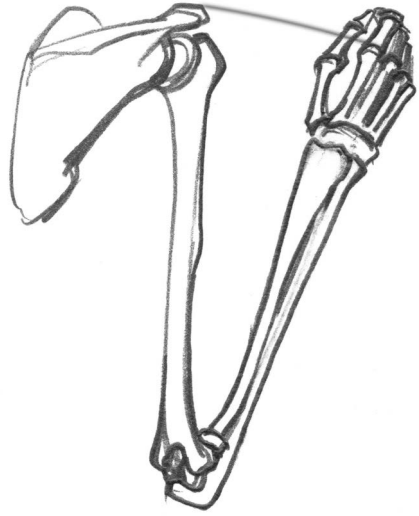
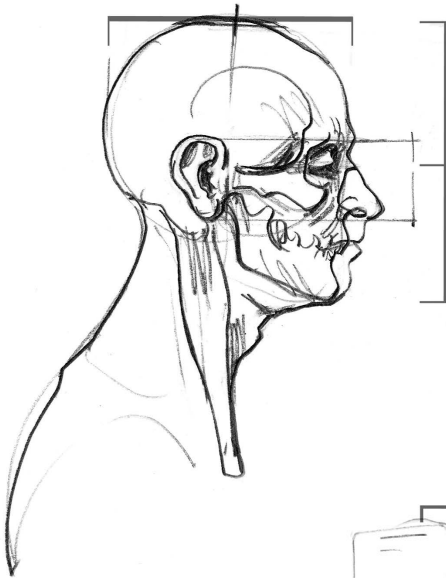
An understanding of the design of the joints and the muscle insertion points should help you to memorize the relationships among the various muscle groups. It will also allow you to understand the mechanics of the human body and to imagine how changes in shape are related to movements (such as stretches, contractions, muscular relaxations, and folds caused by flexion and torsion).

The proportions between the muscles change from one individual to the next, not just in their strength and therefore their thickness, but also in the relationships between their muscle and tendon fibers. A thick muscle will be more powerful. If the muscle fibers are short, the muscle will contract more rapidly.

On the other hand, if the fibers are long, the muscle will be more elastic.

Body Proportions

I take inspiration from some of the canonical works on proportions, including those of Leonardo da Vinci and Paul Richer. However, we need to question these canonical formulations every time a new model stands before us. The canons are useful as a way to think of the body in terms of simple and memorable measurements, so we can then use this information as a way to identify the singular characteristics of each individual. Following, I present the basic tenets of the body's proportions.



Head

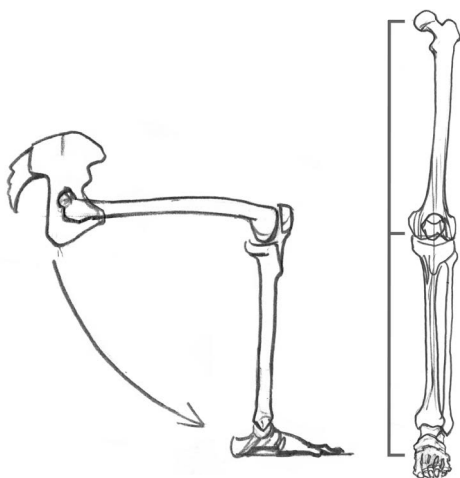
The eyes are situated halfway up the height of the head. This proportion varies with the proportion of the jaw. In a frontal view, the two eyes are separated from each other by the width of one eye.

The ear is level with the nose and situated behind the inferior maxillary joint. Take care to confirm the distance between the ear and the wing of the nose; the most common mistake is to underestimate that distance.

Upper Limb

The distance between the top of the shoulder (the clavicle) and the elbow is equivalent to the distance between the elbow and the heads of the metacarpal bones (the closed fists).

The hand can be separated into two equal segments: from the wrist to the heads of the metacarpals, and from the heads of the metacarpals to the end of the middle finger. On every finger, the length of the first phalange is equal to the combined lengths of the two following phalanx.

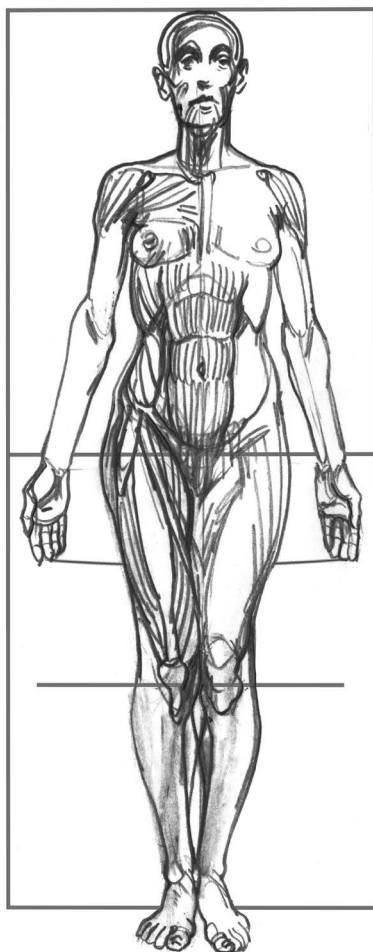


Lower Limb

The halfway point from the hip joint to the ground is at the level of the knee joint. Indeed, if the leg is flexed, the heel will be positioned under the buttocks.

Overview

Da Vinci's canonical drawing—which depicts a standing man with his arms spread, inside a circle within a square (see page 10)—highlights a proportional relationship that is useful for drawing: our height corresponds to our wingspan (the distance between the ends of our two hands when our arms are outstretched



and aligned). Richer, using multiple measurements and averages, expanded on this rule: a man's wingspan most often exceeds his height, while woman's height most often exceeds her wingspan. The difference between the widths of a man's and a woman's shoulders, which are included in the measurement of the wingspan, explains this discrepancy.

The halfway point of the total height when in a standing position can be found at the pubis, and the quarter point is at the knee joint. With the arms straight down along the body, the end of the hand is at the level of the mid-thigh. The elbow is at the level of the waist.



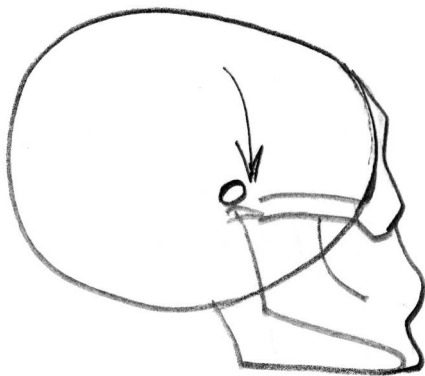
Balance (Plumb Lines)

Using a plumb line (a simple object hung from the end of a string can do the trick) or a vertical line in your model's environment (a wall angle, door frame, etc.), you will be able to check the alignment of various points on the body and the balance between the different parts. Attention to the plumb line will make it possible to render the weight of the body, its stability or alternatively its imbalance, and the dynamics of a pose.

In profile—with the model in a standing position and with the arms hanging along the side of the body—a vertical line will fall along

the ear canal (and the lower jaw joint) and will pass in front of the shoulder, in back of the hip joint, in front of the knee joint, and onto the top of the plantar arch. The spinal column connects with the skull along this plumb line, supporting it from below by curving forward. The spine remains behind the line as far down as the lumbar vertebrae. At the small of the back the vertebrae curves to pass in front of the centerline before rejoining the sacrum behind the line again.

Here we see the front view of a leg while standing, with a vertical line dropped from the hip joint (at the level of the head of the femur).



The line will pass through the middle of the knee and ankle joints. Note the direction of the femur: the neck of the femur pushes it to the outside and forces it to rejoin the vertical line at the knee by descending at an angle.

Head and Neck

Bone Reference Points

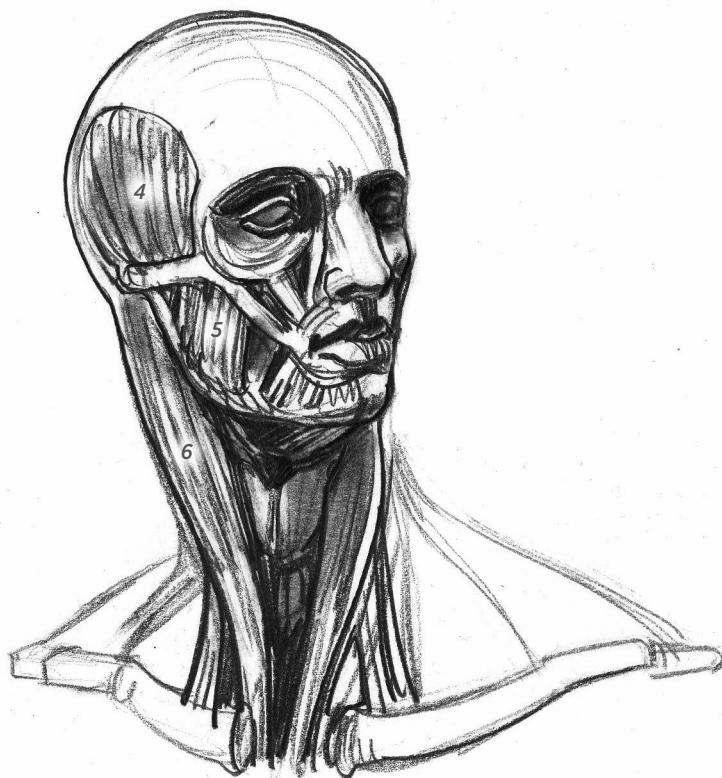
In profile, a skull can be drawn as connecting the ovoid cranial box with the facial bones. The facial bones include the perimeter of the eye sockets, the cheekbones, and the lower jaw, forming a circular arc that curves back up to the ears.

The ear canal is an important reference point for the construction of the skull. Situated at the mid-width of the profile, it is the



meeting point between the lower jaw and the beginning of the zygomatic bone, which runs along under the skin up to the cheekbone. You will be able to position the eye sockets by referring to the proportional rule that, in a frontal view, puts the eyes at mid-height in the face (but remember to reevaluate the position on your individual model).

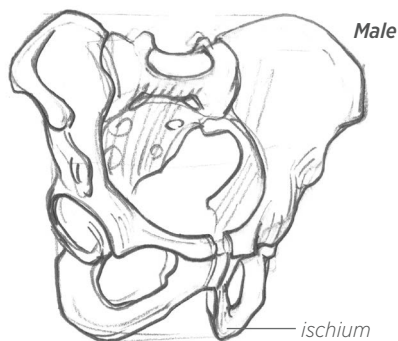
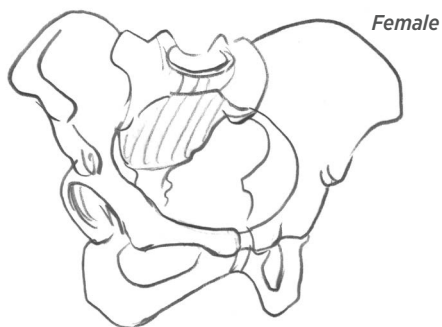
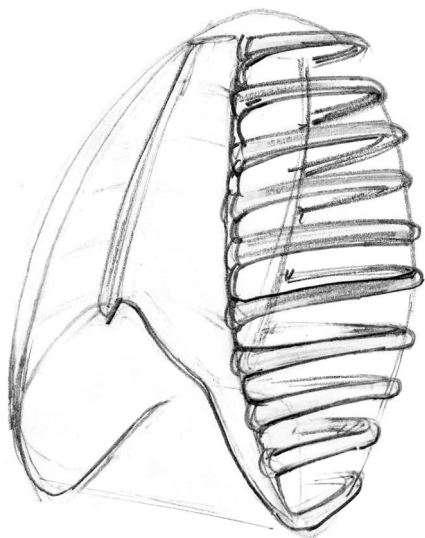
The Adam's apple, or thyroid cartilage (thy – see front cover flap), influences the shapes in the neck area. It has the appearance, under the skin, of a bone reference point. This cartilage, which is suspended from the lower jaw by a little horseshoe-shaped bone called the hyoid bone, then joins the trachea, which is supported on the axis of the breastbone (the sternum).



Fleshed Forms

The musculature of the head includes two masticatory muscles that can influence shape: these are the temporal muscle (4) and the masseter muscle (5). Both are connected to the lower jaw. On the face, these muscles are attached to the skin and blend with it. Because an *écorché* figure has, by definition, been “undressed” of the skin, it is difficult to describe their impact. These muscles mostly radiate out from the mouth. I don’t think that it’s necessary to know their insertion points; here we will only address their formation, around and within the thickness of the lips.

On the neck, it is advisable to first draw the position the sternocleidomastoid muscles (6), which are highly effective in creating structure. These expressive muscles become vertical when one turns one’s head. They make up the connection between the skull and the rib cage, drawing two beautiful lines that follow the Adam’s apple. The thyroid gland is positioned under the Adam’s apple and can round, or soften, the design of this area, especially among women.



Torso

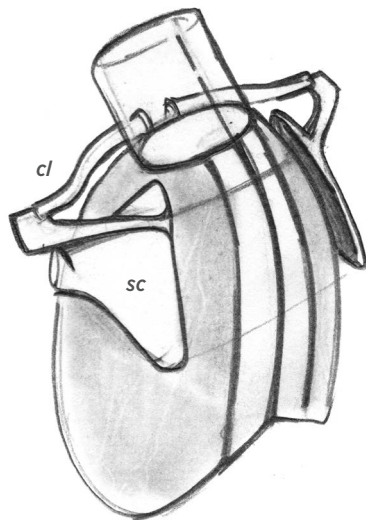
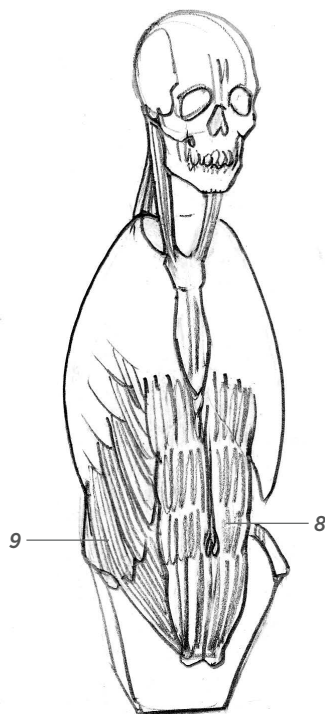
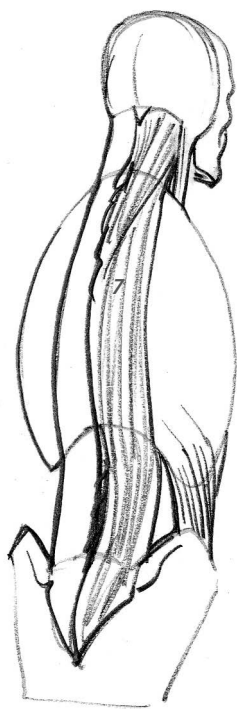
Bone Reference Points

The rib cage has an ovoid shape. Its widest part is usually found just above the waist, and it gently narrows below this level. The space separating the rib cage from the pelvis is smaller than it appears (three finger widths, on average). In certain positions (e.g., bending), the rib cage can actually be positioned inside the pelvis.

The pelvis girdles the lower abdomen and distributes the weight of the upper body across the two femurs, by way of the pelvic ring. The circumference of this ring (about a

third of which involves the sacrum) and the space separating the two ischium bones (the lowest projections on the pelvic girdle), are ways of distinguishing between the male and female pelvis.

In terms of absolute size, the male pelvis is wider. However, in respect to overall body mass, the female pelvis is wider, relatively. The wings and arches that develop above and below the ring provide potential insertion surfaces for the muscles of the abdominal belt and of the thigh.



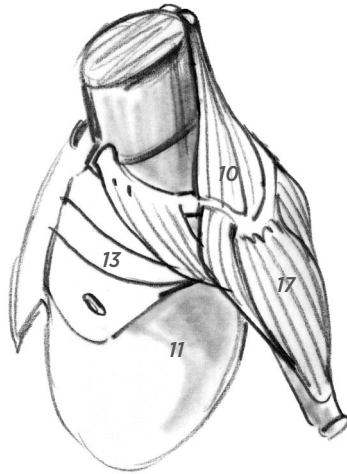
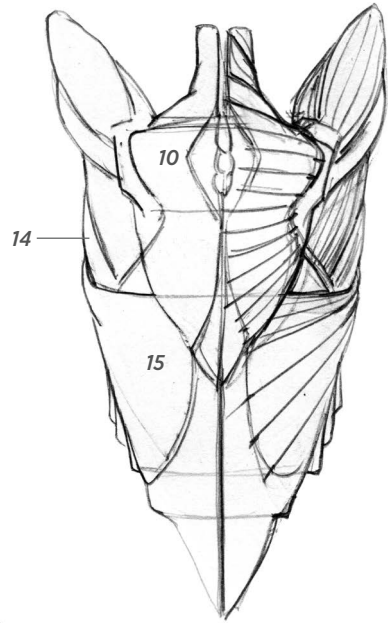
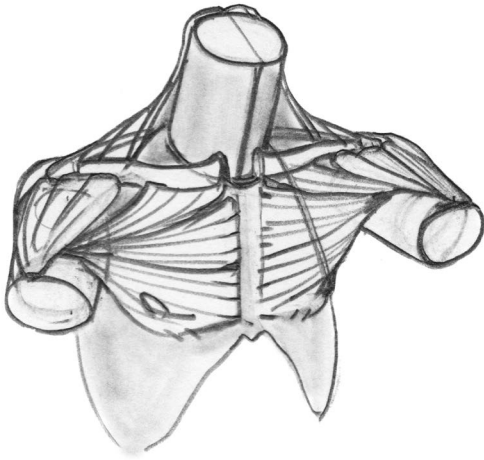
Fleshed Forms

The spinal muscles (7)—the set of muscles that run all along the spine, from the sacrum to the skull—are deep muscles, covered by a superficial musculature, but they are thick and visible enough to deserve our attention. Along with the rectus abdominis muscles (8, abdominals) and the large oblique muscles (9), they constitute a muscular abdominal belt, and together they make up the torso's musculature. These torso muscles, functionally, constitute the “roots” of the arm.

Roots of the Upper Limb

Bone Reference Points

The scapula (sc) and clavicle (cl) can be considered the first bones of the upper limb (i.e., the arm). In terms of mechanics, their purpose is to allow the arm to make certain movements. Every change in position of the arm will result in modifications, often dramatic ones, in the drawing of the torso. In theory, if the scapula were rigidly attached to the rib cage, we would not be able to lift our arm any higher than the horizontal. The scapula has to tilt and turn upwards in order for the arm to be able to lift fully.

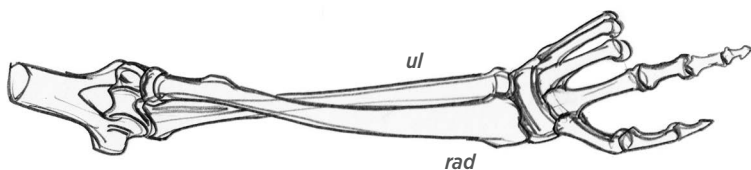
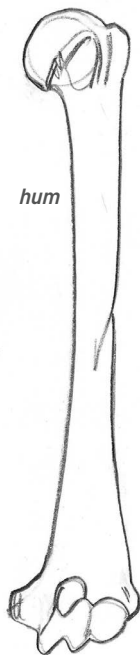


The comparison between human and animal anatomy (comparative anatomy) gives us information about the usefulness of the clavicle, which is absent in many mammals (but that absence promotes other skills), and thus we can note how the musculature of our upper limbs, in combination with our shoulder girdle (scapulas and clavicles), makes it possible for the arm to lift in every direction.

Fleshed Forms

The roots of the arm correspond to the pectoral (13), teres major (14), large dorsal (15), trapezius (10), and serratus anterior muscles. The first three are primarily for lowering the arm and therefore form the walls of the armpit, while the last two, by causing the scapula to tilt and pointing it upwards (relayed by the deltoid, 17) make complete elevation possible.

The breast can be considered as a fat pocket that contains the mammary gland. Kept in a fold of skin, it is thus, so to speak, hooked to the clavicle, and will change shape as it follows the clavicle's movements.



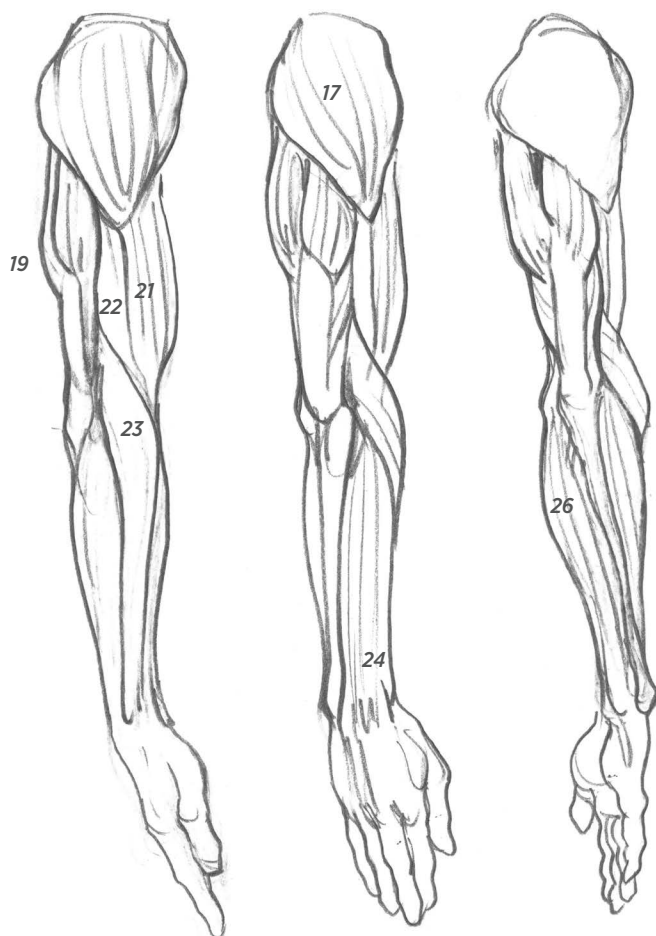
Upper Limb

Bone Reference Points

Even though it is covered by the deltoid, the upper end of the humerus is an important bone reference point. Placed under the connecting point of the clavicle and scapula, it creates a plump form under the muscle.

In the elbow region, this bone ends in a double joint: a sphere attached to a pulley. The sphere is joined to the radius (rad) and is what makes the hand's rotational movements possible. The pulley is connected to the ulna (ul) and allows for movements of flexion and extension. On an écorché of the elbow, the ulna is outlined down as far as the rounded projection that is found near the wrist, on the same side as the little finger.

The shape of the back of the hands and fingers is essentially a result of the skeleton. The small wrist bones are very similar to each other except for two of them that can be found on the "heel" of the hand. One can find a spherical joint (a rotation) on the heads of the metacarpals at the end of the closed fist. But it is the pulleys (for flexion and extension) that separate the different phalanges from each other. Note that there are two phalanges on the thumb and three on each of the other fingers.

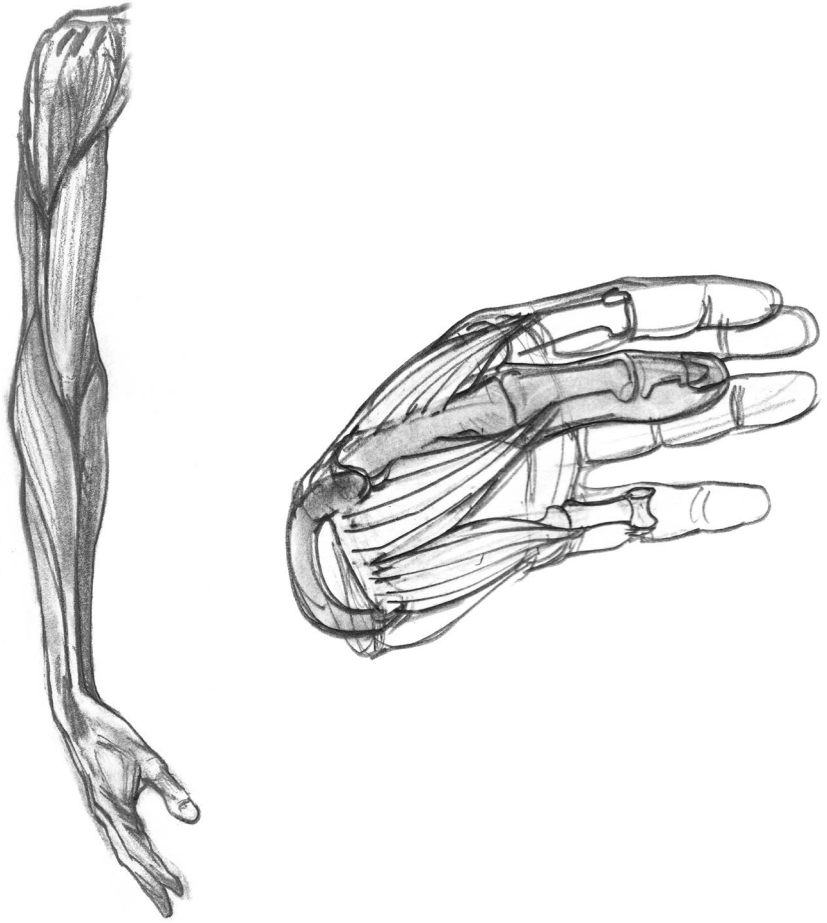


Fleshed Forms

The deltoid muscle (17) caps the shoulder joint and takes part in the elevation of the arm. It has a triangular shape, and its downward-pointing tip is often masked by a fatty mass that extends and is mistaken for the triceps (19).

We will contrast the three bundles of muscle that make up the triceps (three bundles that connect with a shared tendon) with the biceps (21, which has two muscle bundles that share a tendon), the brachial muscle (22, which I will not always show in my drawings for the sake of simplicity), and the brachioradialis muscle (23). The triceps are the extensors for the forearm, while the others are flexors.

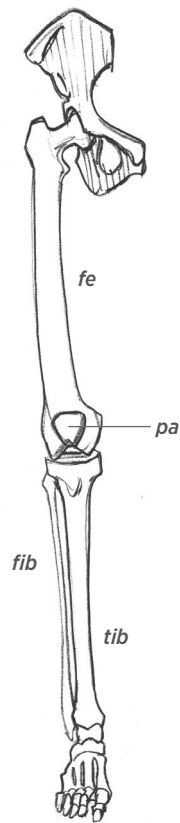
The simplified version of the forearm includes three sets of muscles: to the brachioradialis muscle already mentioned (most often confused with the first radial), we should add the extensors (24, for the hand and the fingers, including the thumb), and the flexors (26, for the hand and the fingers). The so-called “anatomical” position (with the hand open and the thumb facing out in a front view) has often been featured in anatomical works, because it makes it possible to see the bones of the hand and of the forearm in a flat plane, whereas, if the thumb is brought back to the inside, the bones cross over. In order to understand the musculature, however, it is better to start from



a natural position: with the hand hanging down along the side of the body, the palm facing the thigh. In this position, the extensors, which hook to the outside of the humerus, are in a straight line with the back of the hand. As for the flexors, starting with the internal extremity of the humerus, they are in a straight line with the palm. Finally, the brachioradial muscle slips in between the two previous groups.

The hand can be compared to a pair of pliers, with a capacity for gripping inscribed in its resting forms. The thumb's capacity to meet the other fingers makes its orientation paramount. This can be rendered if care is given to its volumes, jointed folds, and nails.

The mobility of the thumb and the relative mobility of the little finger require the presence of muscles at their base, which contribute to making the palm fleshy. However, underneath the hand, fat also plays an important role. It protects the bony extremities much like a cat's paw pads. This fat is situated under the heads of the metacarpals and underneath each phalanx. It is what makes the palm appear longer than the back of the hand. The heads of the metacarpals, which are uncovered on the back of the hand, are shrouded on the palm. It is this discrepancy between the two sides of the hand that explains the folds on the palms, as well as the beveled design that allows the



two sides of the hand to touch, between the fingers. As for the lines on the hand, they are flexion folds for the thumb (so-called lifelines) and for the fingers (so-called heart lines and head lines).

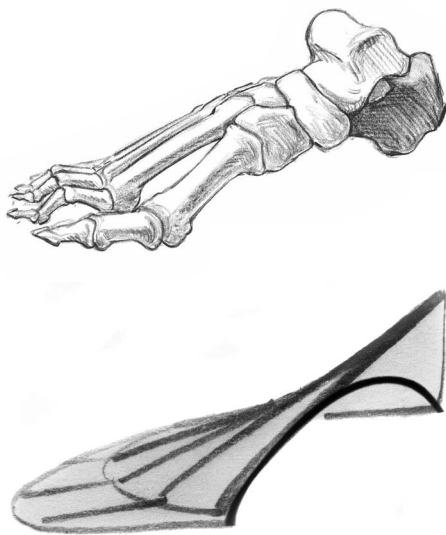
Lower Limb

Bone Reference Points

The pelvis marks the boundary between the torso and the lower limb and is visible at the top, and the ends of its wings are visible underneath the flank. The genitals are outlined underneath the pubis, masked by the fat in that area and, most often, by the pubic hair. At the back, the sacrum (*sacr*) is surrounded by

three bone markers that delineate it under the skin, namely the beginning of each of the two wings of the pelvis, above, and the beginning of the line of the buttocks, below.

The femur (*fe*), at the level of the hip joint when it is not veiled by fat, is outlined under the skin. This bone reappears at the knee, behind the patella (*pa*), a small mobile bone enveloped in the quadriceps tendon. At this level one can easily see the tibia (*tib*) under the skin; it remains underneath the skin down to the ankle (the internal malleolus). The fibula (*fib*) is only visible at its two extremities, and is also a part of the shape of the ankle (the external malleolus).



To fully understand the foot, it is important to be familiar with its bone structure. The foot is designed with a natural arch. Its role is to withstand the body's shocks and weight. The plantar arch takes this dynamic into account inside the foot, as we know, but also on top of the foot (we will not deal with the outer arch here). Thus, the foot is convex, and unless you are dealing with a flat foot, this form should be respected, from the ankle to the heads of the metatarsals.

To this schematic vision, we should add another note: the entire foot skeleton has a spiral shape. The foot, raised up on its inner edge (on the side of the higher malleolus, or ankle

bone), lowers itself gradually toward its outer edge (the lower ankle bone).

Thus, we can distinguish a dynamic part (the internal arc associated with the abductor muscle and the big toe) and a static part (the three last metatarsals and the toes associated with them).

Fleshed Forms

The pelvic wing offers a large surface of insertion into the gluteus medius (31), sometimes called the "deltoid of the hip." The gluteus maximus (32) follows it, inserting itself as far as the sacrum. There are many *écorché* versions of this region. I prefer to accentuate the





33

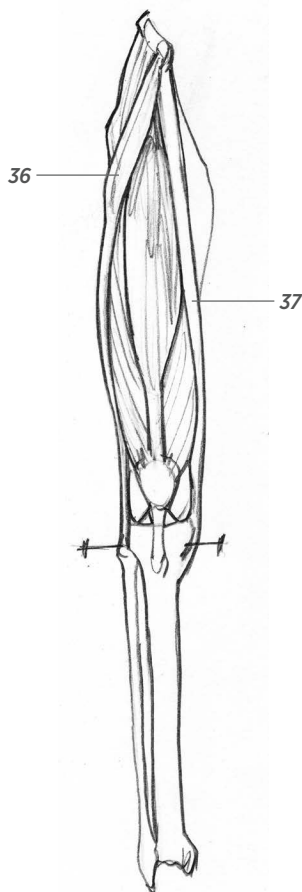
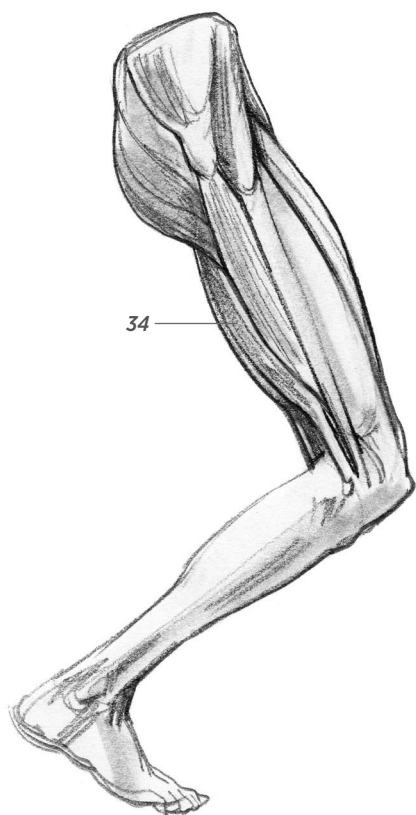


main part of this muscle, which connects directly to the first third of the femur, slipping itself between the quadriceps (33) and the hamstrings (34, next page).

A common mistake is to confuse muscles and fat in the area of the buttocks. The outline of the buttocks is due to fat so it is difficult to evaluate the muscular proportions there. We can only guess at it by relying on the proportions in other parts of the body. At any rate, it seems more interesting to me to try to simplify and synthesize the fatty shapes here. Without the fat, and with the gluteus maximus taking

the most direct path from the sacrum to the femur, we would not have the groove of the buttocks, whose length depends precisely on the thickness of the adipose tissue.

The quadriceps, as the name suggests, are made up of four muscle bundles that meet at a shared tendon, which, after having enveloped the patella, is inserted into the tibia. Of these bundles, we will only retain the three most important. The fourth one, which is a deep one, cannot be drawn. While the two lateral bundles start at the femur, which imposes its direction on the muscle, the central bundle connects to

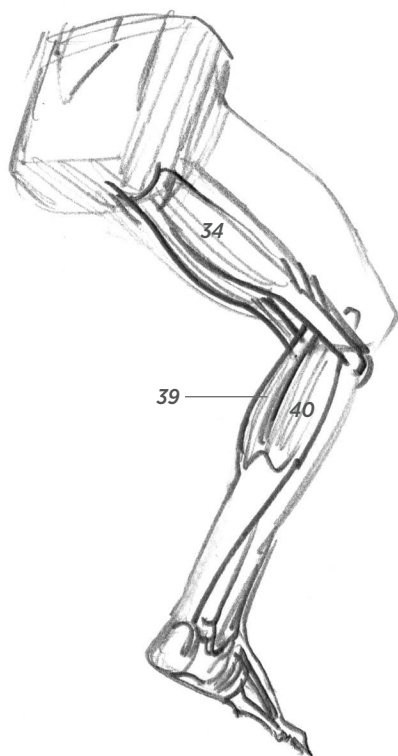


the pelvis and meets the others on the tibia.

The tensor of the fascia lata (36) is a muscle with a very particular mode of insertion. Here, again, I had to make a choice. One has to imagine that the quadriceps, like all muscles, are wrapped in a fibrous "stocking" that contains them and gives them their shape, accompanying its fibers into the depths, all the way to the skeleton. This stocking (the fascia) continues up the leg, wrapping the knee in passing. The tensor, when it is contracted (flexed thigh) or tensed (stationary hip), will tend to stretch the fascia and to give it the shape of a long ribbon,

which can be seen along the side of the quadriceps and of the knee. The tensor of the fascia lata and the sartorial muscle (37) seem to work together, enclosing the thigh before connecting with the knee under the joint, as though to reinforce it on both sides. Their layout vaguely resembles the reins of a harness connecting with a jaw bit at the level of the knee.

I also propose a simplified version of the hamstrings. We will reduce these muscles to two long spindles, which, starting from the ischium (the lowest point of the pelvis), move apart as they descend, allowing the



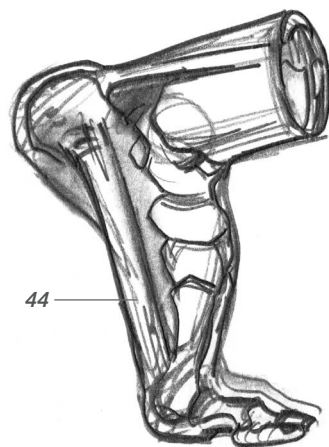
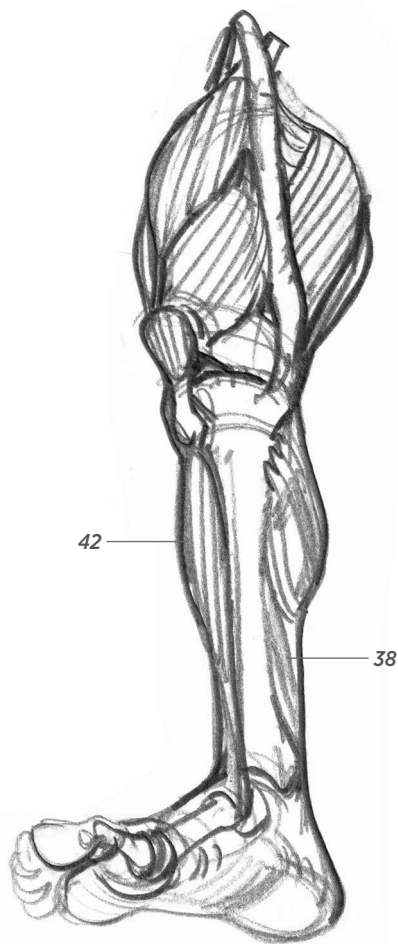
gemelli (39 and 40), behind the knee, to slip between them. When the leg is flexed, the hamstrings (34) end in two long, beautiful tendons on each side of the hollow at the back of the shin.

The simplified version of the adductors (35) involves considering only the strongest bundles and giving them all just one name. This is especially true because these muscles are most often hidden under a thick level of fat that starts at the beginning of the thighs and narrows as it moves downwards.

There is a symmetrical fatty spot on the other side of the thigh, under the hip joint; this is

a typically feminine configuration. If we compare the lower and upper limbs, this fatty spot is reminiscent of the one that can be found behind the tip of the deltoid. In general, a simplified way of describing this is to say that the fatty layer becomes smaller as it moves from the root to the extremities of the limbs.

The gemelli and the soleus muscle (38), taken together, form a triceps. Their shared tendon is none other than the Achilles, which inserts itself into the heel (calcaneus), forming a powerful lever. In the front of the leg, the extensors (42) are reminiscent of those of the forearm: in both cases, the bundles



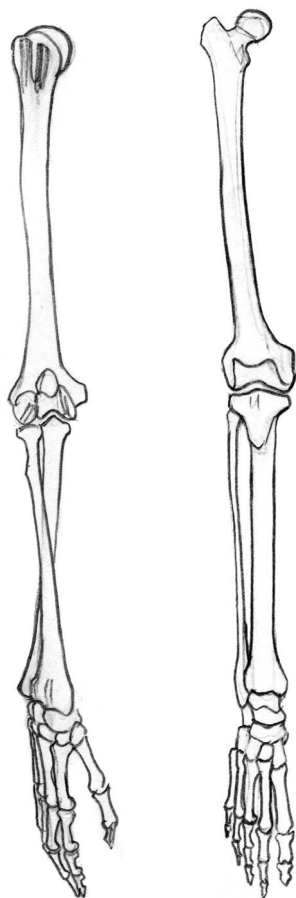
start on the outside and then rotate as they move down, to insert into the back; in one case the back of the foot, in the other the back of the hand.

The foot is designed like a bow, with the skeleton corresponding to the wood of the bow and the “string” being the abductor muscle (44) of the big toe. This construction of the foot can be seen in the footprint it leaves on the ground. But that footprint can also mislead us about the strength of the foot’s dynamics, because the arch may be masked by the fat in that area. This fat is reminiscent of the fat on the hand, and obvi-

ously is involved in the foot’s shock-absorbing role, providing a veritable mattress atop the bones. As on the hand, we find the skin folds and the beveled plane between the toes, for the same reasons.

Similarities Between Arms and Legs

The task here is to compare the skeleton of the upper and lower human limbs. For certain animals, we could take the comparison as far as the scapular (scapula, clavicle) and pelvic girdles, but for the purposes of this book, that will not be necessary.



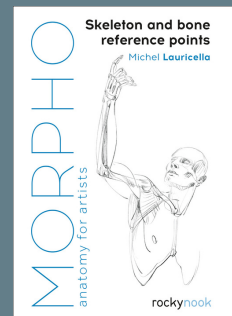
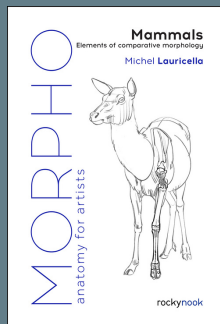
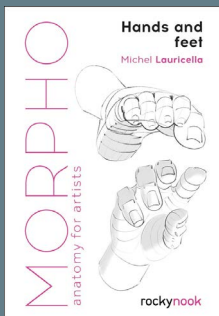
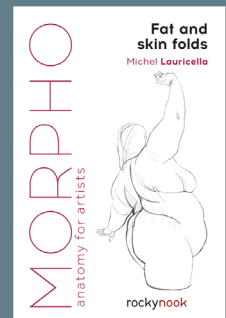
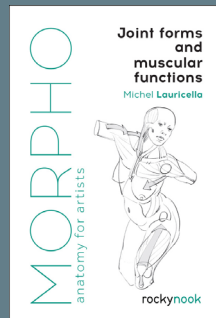
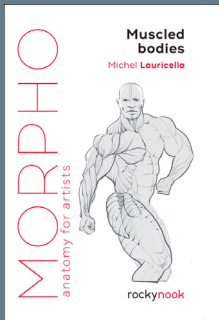
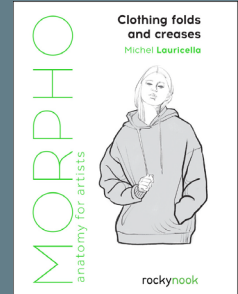
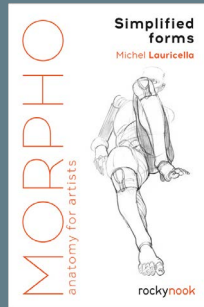
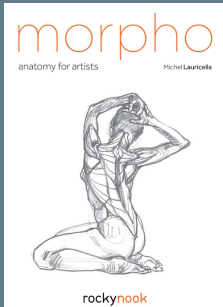
On both limbs, the lower and the upper, there is only one bone on the first segment: the humerus and the femur, respectively. Then, on the second segment, there are two bones: the radius and the ulna on the arm, and the tibia and the fibula on the leg. Then come the small bones of the wrist, which could correspond to those of the instep. Finally, there are five fingers on the hand, corresponding to the five toes on the foot, with the same number of phalanges on each extremity, and in both cases one fewer on the thumb and on the big toe.

As we saw earlier, the muscles that operate the hand and the foot are inserted above the wrist and ankle joints, and rest on the paired bones of the forearm and leg, respectively.

It would seem that the complexity of the extremities, in particular the number of digits (and here a comparison with the anatomies of various mammalian species is very informative), determines and necessitates the enlargement (on the same plane as the hand or foot) of the insertion zones. It is the paired juxtaposition of the bones, the radius and ulna on one hand and the tibia and fibula on the other, that makes this possible.

The simplest formation is the one we find on the lower limb (with the tibia and fibula always remaining parallel). On the forearm, the ability to cross and uncross the two bones facilitates rotation and increased movement in the hand such as pronation and supination.

Collect The Entire Morpho Series



<https://rockynook.com/Morpho>